

Digital SAT Math

Topic 2: Exponential Functions - Growth and Decay

Complete Solutions & Answer Key

Total: 60 Questions

Format: Digital SAT Style — MCQ & Grid-In

Difficulty: Easy (Q1-20) • Medium (Q21-45) • Hard (Q46-60)

Q1. Correct Answer: C

STEP-BY-STEP SOLUTION:

$$f(4)=3(2)^4=3(16)=48$$

WHY EACH OPTION IS WRONG:

A) $24=3(2)^3=f(3)$

B) 30: 10 is not a power of 2

D) $96=3(2)^5=f(5)$

TRAP ALERT: Computing $3 \cdot 2 \cdot 4 = 24$ (linear) instead of $3 \cdot 2^4 = 48$ (exponential).

KEY INSIGHT: Evaluate the exponent first, then multiply by coefficient.

FAST METHOD: $2^4=16$; $3 \cdot 16=48$.

Q2. Correct Answer: B

STEP-BY-STEP SOLUTION:

Doubles every hour: multiply by 2 each hour. After t hours: $500(2)^t$

WHY EACH OPTION IS WRONG:

- A) $500+2t$ is linear
- C) $500 \cdot 2t$ means $1000t$ (linear)
- D) Coefficient and base swapped

TRAP ALERT: Writing $500 \cdot 2t$ (linear multiplication) vs $500(2)^t$ (exponential).

KEY INSIGHT: Exponential: initial * (growth factor)^{time}. Variable must be in the EXPONENT.

FAST METHOD: Start * base^t = $500(2)^t$.

Q3. Correct Answer: B

STEP-BY-STEP SOLUTION:

Base=0.5. Since $0 < 0.5 < 1$, g DECREASES as x increases.

WHY EACH OPTION IS WRONG:

- A) Base >1 needed for growth
- C) Coefficient affects starting value, not direction
- D) Large coefficient doesn't cause growth

TRAP ALERT: Confusing the COEFFICIENT (4) with the BASE (0.5). Base determines growth vs. decay.

KEY INSIGHT: For ab^x : if $b > 1$ then growth; if $0 < b < 1$ then decay. Coefficient is irrelevant.

FAST METHOD: $b = 0.5 < 1 \rightarrow$ decay.

Q4. Correct Answer: C

STEP-BY-STEP SOLUTION:

At $t=0$: $V(0) = 18,000(0.85)^0 = 18,000(1) = \$18,000$

WHY EACH OPTION IS WRONG:

- A) \$0.85 is the decay factor
- B) $\$15,300 = V(1)$, value after 1 year
- D) Would imply appreciation

TRAP ALERT: Confusing the coefficient (initial value) with the decay rate.

KEY INSIGHT: In $V = a \cdot b^t$: coefficient a equals value at $t=0$ since $b^0 = 1$.

FAST METHOD: $t=0 \rightarrow b^0 = 1 \rightarrow V = a = \$18,000$.

Q5. Correct Answer: C

STEP-BY-STEP SOLUTION:

$f(x)=3(1.2)^x$: base= $1.2>1 \rightarrow$ exponential GROWTH.

WHY EACH OPTION IS WRONG:

- A) $3x+5$ is linear
- B) x^3 is polynomial (x is base, not exponent)
- D) Base $0.8<1 \rightarrow$ decay

TRAP ALERT: Confusing polynomial (x is base) with exponential (x is exponent).

KEY INSIGHT: Exponential growth: form $a \cdot b^x$ with $b>1$. Variable **MUST** be in exponent.

FAST METHOD: Base $1.2>1 \rightarrow$ growth.

Q6. Correct Answer: D

STEP-BY-STEP SOLUTION:

$h(0)=100(0.9)^0=100(1)=100$

WHY EACH OPTION IS WRONG:

- A) 0: function has nonzero initial amount
- B) 0.9 is the base
- C) $90=100(0.9)=h(1)$

TRAP ALERT: Using the base value (0.9) instead of evaluating at $t=0$.

KEY INSIGHT: Any nonzero base to the power 0 equals 1, so $f(0)=\text{coefficient}$.

FAST METHOD: $b^0=1 \rightarrow h(0)=100$.

Q7. Correct Answer: D

STEP-BY-STEP SOLUTION:

$f(0)=5(3)^0=5(1)=5$. $f(1)=5(3)^1=15$. Sum= $5+15=20$

WHY EACH OPTION IS WRONG:

- A) 8: arithmetic error
- B) 10: confusing $f(0+1)=f(1)$ with $f(0)+f(1)$
- C) $15=f(1)$ alone

TRAP ALERT: Computing $f(0+1)=f(1)=15$ instead of $f(0)+f(1)=5+15=20$.

KEY INSIGHT: $f(0)+f(1)$ means evaluate at each input separately, then add.

FAST METHOD: $5+15=20$.

Q8. Correct Answer: B

STEP-BY-STEP SOLUTION:

4% growth, compounded annually. Growth factor = $1 + 0.04 = 1.04$. $A(t) = 1,000(1.04)^t$

WHY EACH OPTION IS WRONG:

- A) Simple (linear) interest
- C) $(0.04)^t$ makes value nearly 0 instantly
- D) $(4)^t$ quadruples each year

TRAP ALERT: Using 0.04 (the rate) as the base instead of 1.04 (the growth factor).

KEY INSIGHT: Compound growth: multiply by $(1+r)$ each period. For 4%: factor = 1.04.

FAST METHOD: $A(t) = P(1+r)^t = 1,000(1.04)^t$.

Q9. Correct Answer: B

STEP-BY-STEP SOLUTION:

Half-life $T=10$ years. After t years: $Q(t) = 80(0.5)^{(t/10)}$

WHY EACH OPTION IS WRONG:

- A) $80-4t$ is linear
- C) $80(2)^t$ grows (wrong direction)
- D) $80(10)^t$ grows explosively

TRAP ALERT: Using t directly as exponent instead of t/T (number of half-lives).

KEY INSIGHT: Half-life formula: $Q = Q_0(0.5)^{(t/T)}$. Exponent counts half-lives elapsed.

FAST METHOD: $T=10 \rightarrow$ exponent is $t/10$.

Q10. Correct Answer: B

STEP-BY-STEP SOLUTION:

$f(0) = a = 6$. $f(1) = 6b = 18 \rightarrow b = 3$

WHY EACH OPTION IS WRONG:

- A) $b=2$: $f(1) = 12 \neq 18$
- C) $b=6$: $f(1) = 36 \neq 18$
- D) $b=12$: $f(1) = 72 \neq 18$

TRAP ALERT: Confusing b with $f(1)$; b is the RATIO $f(1)/f(0)$.

KEY INSIGHT: Use $f(0)$ to find a ; then $b = f(1)/a$.

FAST METHOD: $b = 18/6 = 3$.

Q11. Correct Answer: B

STEP-BY-STEP SOLUTION:

20% decrease: remaining factor= $1-0.20=0.80$. After 3 years: $800(0.8)^3$

WHY EACH OPTION IS WRONG:

- A) Subtracts fixed 20% of original each year (linear)
- C) $800(1.2)^3$: 20% increase
- D) $800(0.2)^3$: uses rate as base (nearly eliminates value)

TRAP ALERT: Subtracting a fixed amount yearly (linear) vs multiplying by 0.8 (exponential).

KEY INSIGHT: Compound decay: multiply by $(1-r)$ each period. $1-0.20=0.80$.

FAST METHOD: Decay factor= $0.80 \rightarrow 800(0.8)^3$.

Q12. Correct Answer: B

STEP-BY-STEP SOLUTION:

$p(t)=200(1.05)^t$: t is in the exponent $\rightarrow$ exponential. Base= $1.05>1 \rightarrow$ growth at 5% per period.

WHY EACH OPTION IS WRONG:

- A) Linear: form $mt+b$, not ab^t
- C) Decay needs base <1
- D) Quadratic has t^2

TRAP ALERT: Seeing coefficient 200 and thinking it's a slope of a linear function.

KEY INSIGHT: Variable in the EXPONENT position = exponential function.

FAST METHOD: t in exponent $\rightarrow$ exponential; base $1.05>1 \rightarrow$ 5% growth.

Q13. Correct Answer: C

STEP-BY-STEP SOLUTION:

y-intercept: $f(0)=7(4)^0=7(1)=7$. Point: $(0,7)$

WHY EACH OPTION IS WRONG:

- A) $(0,1)$: coefficient would need to be 1
- B) $(0,4)$: coefficient would be 4
- D) $(0,28)=f(1)$, not y-intercept

TRAP ALERT: Evaluating $f(1)$ instead of $f(0)$ for the y-intercept.

KEY INSIGHT: y-intercept always at $x=0$. $f(0)=a*b^0=a$.

FAST METHOD: $f(0)=7$.

Q14. Correct Answer: C

STEP-BY-STEP SOLUTION:

3% growth: multiply by 1.03. After 1 year: $50,000(1.03)$

WHY EACH OPTION IS WRONG:

- A) Adding 0.03 is meaningless (units mismatch)
- B) $50,000(0.03)=1,500$ (a small fraction)
- D) $50,000(3)$ triples population

TRAP ALERT: Multiplying by 0.03 (rate) instead of 1.03 (growth factor).

KEY INSIGHT: For $r\%$ growth: $\text{new value} = \text{old} * (1+r/100) = \text{old} * 1.03$.

FAST METHOD: Growth factor=1.03 $\rightarrow 50,000*1.03$.

Q15. Correct Answer: B

STEP-BY-STEP SOLUTION:

$$g(x)=2^{(x+3)}=2^x * 2^3=8*2^x=8*f(x)$$

WHY EACH OPTION IS WRONG:

- A) $g(x)=f(x)+3$: shifts graph up 3, gives 2^x+3
- C) $g(x)=f(x)-3$: shifts down 3
- D) $f(x)/8$ would require $2^{(x-3)}$

TRAP ALERT: Adding 3 to the exponent looks like it adds 3 to the output, but it multiplies by $2^3=8$.

KEY INSIGHT: $b^{(x+k)}=b^k * b^x$. Adding k to exponent multiplies function by b^k .

FAST METHOD: $2^{(x+3)}=8*2^x=8*f(x)$.

Q16. Correct Answer: A

STEP-BY-STEP SOLUTION:

Ratios: $3/1=3$, $9/3=3$, $27/9=3$. Constant ratio $\rightarrow$ exponential.

WHY EACH OPTION IS WRONG:

- B) Differences 2,6,18 are NOT constant
- C) Constant differences $\rightarrow$ linear, not exponential
- D) Exponential y-intercept can be any positive value

TRAP ALERT: Confusing constant RATIO (exponential) with constant DIFFERENCE (linear).

KEY INSIGHT: Exponential functions have constant multiplicative ratio between consecutive outputs.

FAST METHOD: Ratio=3 each time $\rightarrow$ exponential.

Q17. Correct Answer: 40

STEP-BY-STEP SOLUTION:

$$f(3)=5(2)^3=5(8)=40$$

TRAP ALERT: Computing $5 \cdot 2 \cdot 3 = 30$ (treating as multiplication) instead of $5 \cdot 2^3$.

KEY INSIGHT: Evaluate the exponent ($2^3=8$) before multiplying by coefficient.

FAST METHOD: $5 \cdot 8 = 40$.

Q18. Correct Answer: A

STEP-BY-STEP SOLUTION:

$$d(t)=1,200(0.75)^t. \text{ Base}=0.75=1-0.25 \rightarrow 25\% \text{ decrease per hour.}$$

WHY EACH OPTION IS WRONG:

B) 75% is what REMAINS, not what decreases

C) 80% would mean base=0.80

D) 125% implies growth

TRAP ALERT: Confusing remaining fraction (0.75) with percent decrease (25%).

KEY INSIGHT: If base $b=1-r$, then r is the percent decrease. $1-0.75=0.25=25\%$.

FAST METHOD: $1-0.75=0.25=25\% \text{ decrease}$.

Q19. Correct Answer: A

STEP-BY-STEP SOLUTION:

$$f(2)=6b^2=54 \rightarrow b^2=9 \rightarrow b=3$$

WHY EACH OPTION IS WRONG:

B) $b=6$: $6(36)=216 \neq 54$

C) $b=9$: $6(81)=486 \neq 54$

D) $b=27$: far too large

TRAP ALERT: Not dividing by 6 first: $b^2=54/6=9$, not $b^2=54$.

KEY INSIGHT: Isolate b^x by dividing both sides by a , then take the root.

FAST METHOD: $b^2=9 \rightarrow b=3$.

Q20. Correct Answer: C

STEP-BY-STEP SOLUTION:

Triples every 2 weeks. In 6 weeks: $6/2=3$ periods. Total $=400 \cdot 3^3 = 400 \cdot 27 = 10,800$

WHY EACH OPTION IS WRONG:

- A) $1,200=400 \cdot 3$: only 1 period (2 weeks)
- B) $3,600=400 \cdot 9=400 \cdot 3^2$: 2 periods (4 weeks)
- D) $32,400=400 \cdot 81$: 4 periods (8 weeks)

TRAP ALERT: Treating 6 weeks as 6 tripling periods instead of $6/2=3$ periods.

KEY INSIGHT: Number of periods = total time / period length.

FAST METHOD: 3 periods $\rightarrow 400 \cdot 3^3 = 10,800$.

Q21. Correct Answer: A

STEP-BY-STEP SOLUTION:

$a=f(0)=12$. $f(3)=12b^3=96 \rightarrow b^3=8 \rightarrow b=2$

WHY EACH OPTION IS WRONG:

- B) $b=3$: $12(27)=324 \neq 96$
- C) $b=4$: $12(64)=768 \neq 96$
- D) $b=8$: far too large

TRAP ALERT: Not dividing by $a=12$ before solving for b .

KEY INSIGHT: Use $f(0)$ to find a ; then use another point to find b .

FAST METHOD: $b^3=96/12=8 \rightarrow b=2$.

Q22. Correct Answer: B

STEP-BY-STEP SOLUTION:

Ratios: $10/5=2$, $20/10=2$, $40/20=2$. Constant ratio=2, start=5. Model: $y=5(2)^x$

WHY EACH OPTION IS WRONG:

- A) $y=5x+5$: linear; $f(2)=15 \neq 20$
- C) $y=2(5)^x$: $f(0)=2 \neq 5$
- D) $y=10x$: $f(0)=0 \neq 5$

TRAP ALERT: Option C matches $x=1$ but fails at $x=0$.

KEY INSIGHT: Always check y-intercept ($x=0$) and at least one other point.

FAST METHOD: y-intercept=5, ratio=2 $\rightarrow y=5(2)^x$.

Q23. Correct Answer: B

STEP-BY-STEP SOLUTION:

$$3,000(1+r)^2=3,363 \rightarrow (1+r)^2=1.121 \rightarrow 1+r=1.0588 \rightarrow r \approx 0.059 \approx 0.06$$

WHY EACH OPTION IS WRONG:

A) $r=0.03$: $3,000(1.03)^2 \approx \$3,183 \neq \$3,363$

C) $r=0.10$: $3,000(1.10)^2 = \$3,630 \neq \$3,363$

D) $r=0.08$: $3,000(1.08)^2 \approx \$3,499 \neq \$3,363$

TRAP ALERT: Forgetting to SQUARE $(1+r)$; using $(1+r)*2$ instead of $(1+r)^2$.

KEY INSIGHT: After $t=2$ years: $(1+r)^2=3363/3000$. Take square root to find $1+r$.

FAST METHOD: $(1+r)^2=1.121 \rightarrow r \approx 0.06$.

Q24. Correct Answer: B

STEP-BY-STEP SOLUTION:

$$Q(30)=Q_0 * b^{30} = (1/8) * Q_0 \rightarrow b^{30}=1/8 \rightarrow b=(1/8)^{(1/30)}$$

WHY EACH OPTION IS WRONG:

A) $(1/2)^{(1/10)}$: half-life would be 10 min, not 30

C) $(1/2)^{(1/30)} = ((1/8)^{(1/3)})^{(1/30)} = (1/8)^{(1/90)}$: wrong

D) $1/8$: total decay over 30 min, not per-minute factor

TRAP ALERT: Using the total decay fraction $(1/8)$ as the per-minute rate b .

KEY INSIGHT: $b = (\text{fraction remaining})^{(1/\text{time})} = (1/8)^{(1/30)}$.

FAST METHOD: $b^{30}=1/8 \rightarrow b=(1/8)^{(1/30)}$.

Q25. Correct Answer: D

STEP-BY-STEP SOLUTION:

40% increase per unit: multiply by 1.40 each time. So $b=1.4$

WHY EACH OPTION IS WRONG:

A) 0.4: 40% of previous (60% loss!)

B) $0.6=1-0.4$: 40% DECREASE

C) 1.04: 4% increase, not 40%

TRAP ALERT: Using 0.4 as base (means 60% loss!) instead of 1.4 (40% gain).

KEY INSIGHT: 'Increases by r ' $\rightarrow$ multiply by $(1+r)$. For 40%: $b=1.40$.

FAST METHOD: 40% increase $\rightarrow b=1.40$.

Q26. Correct Answer: C

STEP-BY-STEP SOLUTION:

$g(x)=500(1.06)^{(x-3)}=f(x-3)$. So $g(x)=f(x-3)$.

WHY EACH OPTION IS WRONG:

- A) $f(x)-3$: subtracts from output, not exponent
- B) $f(x)/f(3)$: missing the 500 coefficient
- D) $f(x)*f(3)$: would give $500^2*(1.06)^{(x+3)}$

TRAP ALERT: Thinking subtraction in exponent means subtracting from the function value.

KEY INSIGHT: $f(x-c)=500(1.06)^{(x-c)}$: horizontal shift right by c.

FAST METHOD: Replace x with x-3 in f $\rightarrow g(x)=f(x-3)$. ✓

Q27. Correct Answer: D

STEP-BY-STEP SOLUTION:

$N(12)=2,500(1.08)^{12}$. $(1.08)^{12}\approx 2.518$. $N\approx 2,500*2.518\approx 6,295\approx 6,326$

WHY EACH OPTION IS WRONG:

- A) 2,500: initial value ($t=0$)
- B) 3,000: near $t=2$
- C) 5,000: near $t=9$

TRAP ALERT: Confusing the time unit; $t=12$ months IS correct here.

KEY INSIGHT: Evaluate at $t=12$ directly. Use $(1.08)^{12}\approx 2.518$.

FAST METHOD: $2500*2.518\approx 6,295$.

Q28. Correct Answer: B

STEP-BY-STEP SOLUTION:

$f(x+2)=4^{(x+2)}=4^x*4^2=16*4^x=16*f(x)$

WHY EACH OPTION IS WRONG:

- A) $f(x)+16$: adds 16 (doesn't multiply)
- C) $f(x)+2$: shifts up 2
- D) $2*f(x)$: would need $4^{(x+1/2)}$

TRAP ALERT: Thinking +2 in exponent adds 2 to the output value.

KEY INSIGHT: $b^{(x+k)}=b^k*b^x$. Adding k to exponent multiplies by $b^k=4^2=16$.

FAST METHOD: $4^{(x+2)}=16*4^x=16*f(x)$.

Q29. Correct Answer: B**STEP-BY-STEP SOLUTION:**

Half-life $T=4$ days. After d days: $d/4$ half-lives occur. $m=640(0.5)^{(d/4)}$

WHY EACH OPTION IS WRONG:

- A) $640(0.5)^d$: treats every day as a half-life
- C) $640(0.5)^{(4d)}$: far too many halvings
- D) $640-80d$: linear decay

TRAP ALERT: Using d as exponent directly instead of $d/4$.

KEY INSIGHT: For half-life T : $m=m_0(0.5)^{(t/T)}$.

FAST METHOD: $T=4 \rightarrow \text{exponent}=d/4$.

Q30. Correct Answer: C**STEP-BY-STEP SOLUTION:**

$f(1)=ab=10$, $f(4)=ab^4=80$. Divide: $b^3=8 \rightarrow b=2$. Then $a=10/2=5$

WHY EACH OPTION IS WRONG:

- A) $a=2$: $ab=4$, $b=5$; $f(4)=2(625)=1250 \neq 80$
- B) $a=4$: $b=2.5$; $f(4) \neq 80$
- D) $a=8$: $b=1.25$; $f(4) \neq 80$

TRAP ALERT: Solving for a without eliminating it first via the ratio.

KEY INSIGHT: Divide $f(4)$ by $f(1)$ to cancel a : $b^3=8$. Then $a=10/2$.

FAST METHOD: $b^3=8 \rightarrow b=2 \rightarrow a=5$.

Q31. Correct Answer: C**STEP-BY-STEP SOLUTION:**

$y=2^{x-8}$: the -8 is outside the exponential $\rightarrow$ shift DOWN 8 units.

WHY EACH OPTION IS WRONG:

- A) Shift right: $y=2^{(x-8)}$
- B) Shift left: $y=2^{(x+8)}$
- D) Vertical stretch: $y=8 \cdot 2^x$

TRAP ALERT: Subtracting 8 from output (vertical shift) vs. subtracting from input (horizontal shift).

KEY INSIGHT: $f(x)+k$: vertical shift. $k<0 \rightarrow$ down.

FAST METHOD: -8 outside exponent $\rightarrow$ shift down 8.

Q32. Correct Answer: A

STEP-BY-STEP SOLUTION:

Same base (1.04) $\rightarrow$ same growth rate. $g(0)=200+100=300$ vs $f(0)=200$. Higher initial value.

WHY EACH OPTION IS WRONG:

- B) Adding 100 doesn't change the base or growth rate
- C) g grows (base >1); higher starting point, not decay
- D) g is still transformationally exponential in character

TRAP ALERT: Thinking adding 100 changes the growth rate from 4%.

KEY INSIGHT: Adding a constant shifts vertically; doesn't change the multiplicative growth factor.

FAST METHOD: Same base=same rate; $g(0)=300>f(0)=200$.

Q33. Correct Answer: B

STEP-BY-STEP SOLUTION:

$12,000(1.025)^t > 15,000 \rightarrow (1.025)^t > 1.25 \rightarrow t > \ln(1.25)/\ln(1.025) \approx 9.04$. First integer $t=10 \rightarrow$ year 2020.

Verify: $P(9) \approx 14,987 < 15,000$; $P(10) \approx 15,361 > 15,000$ ✓

WHY EACH OPTION IS WRONG:

- A) 2019: $t=9$ gives $P \approx 14,987 < 15,000$ (not yet exceeded)
- C) 2021: one year late
- D) 2022: two years late

TRAP ALERT: Rounding 9.04 down to 9 without verifying $P(9)$ is still below 15,000.

KEY INSIGHT: Solve inequality for t ; round UP and verify.

FAST METHOD: $t > 9.04 \rightarrow$ first integer $t=10 \rightarrow$ year 2020.

Q34. Correct Answer: C

STEP-BY-STEP SOLUTION:

$y=2(1.1)^x$: base=1.1 $>1 \rightarrow$ exponential GROWTH, not decay.

WHY EACH OPTION IS WRONG:

- A) base=0.3 $<1 \rightarrow$ decay
- B) base=0.75 $<1 \rightarrow$ decay
- D) base=0.99 $<1 \rightarrow$ decay

TRAP ALERT: Seeing coefficient 2 (>1) and thinking it means growth when base determines direction.

KEY INSIGHT: Decay: $0 < b < 1$. Growth: $b > 1$. Coefficient is irrelevant to this determination.

FAST METHOD: Base 1.1 $>1 \rightarrow$ growth, NOT decay.

Q35. Correct Answer: B**STEP-BY-STEP SOLUTION:**

$$100e^{(5k)}=200 \rightarrow e^{(5k)}=2 \rightarrow 5k=\ln(2)\approx 0.693 \rightarrow k\approx 0.139\approx 0.14$$

WHY EACH OPTION IS WRONG:

A) $k=0.07$: $e^{(0.35)}\approx 1.42\neq 2$

C) $k=0.20$: $e^{(1.0)}\approx 2.718\neq 2$

D) $k=0.40$: $e^{(2.0)}\approx 7.39\neq 2$

TRAP ALERT: Using doubling time directly as k , rather than $k=\ln(2)/t$.

KEY INSIGHT: $k=\ln(\text{ratio})/\text{time}$. Doubling: $k=\ln(2)/5$.

FAST METHOD: $k=\ln(2)/5=0.693/5\approx 0.14$.

Q36. Correct Answer: B**STEP-BY-STEP SOLUTION:**

$$g(x)=3^{(2x)}=(3^x)^2=[f(x)]^2$$

WHY EACH OPTION IS WRONG:

A) $2*f(x)=2*3^x\neq 3^{(2x)}$

C) $f(x)+f(x)=2*3^x\neq 3^{(2x)}$

D) $f(x+2)=3^{(x+2)}=9*3^x\neq 3^{(2x)}$

TRAP ALERT: Thinking $3^{(2x)}=2*3^x$ (it's actually $[3^x]^2$).

KEY INSIGHT: $(b^x)^2=b^{(2x)}$. So $3^{(2x)}=(3^x)^2=[f(x)]^2$.

FAST METHOD: $3^{(2x)}=[3^x]^2=[f(x)]^2$.

Q37. Correct Answer: B**STEP-BY-STEP SOLUTION:**

$$24,000(0.85)^n < 12,000 \rightarrow (0.85)^n < 0.5 \rightarrow n > 4.27.$$

$n=4$: $24,000(0.85)^4 \approx \$12,528 > \$12,000$ (NOT yet below)

$n=5$: $24,000(0.85)^5 \approx \$10,649 < \$12,000$ ✓

Answer: 5 years

WHY EACH OPTION IS WRONG:

A) After 4 years: $\sim \$12,528$ still above $\$12,000$

C) 6 years: one year later than needed

D) 7 years: two years later

TRAP ALERT: Getting $n > 4.27$ and rounding down to 4 without checking the 4-year value.

KEY INSIGHT: Solve the inequality, then verify at next integer.

FAST METHOD: $(0.85)^4 \approx 0.522 \rightarrow \$12,528 > \$12,000$; $(0.85)^5 \approx 0.444 \rightarrow \$10,649 < \$12,000$. Answer: 5 years.

Q38. Correct Answer: B**STEP-BY-STEP SOLUTION:**

$$f(x)=8(2)^{(x/3)}=8*(2^{(1/3)})^x. \text{ So } b=2^{(1/3)}$$

WHY EACH OPTION IS WRONG:

- A) $b=2$: exponent would need to be x (not $x/3$)
- C) $b=2^3=8$: confuses base and coefficient
- D) $b=8$: that's the coefficient a

TRAP ALERT: Confusing the rewrite: $2^{(x/3)}=(2^{(1/3)})^x$.

KEY INSIGHT: $a^{(bx)}=(a^b)^x$. So $2^{(x/3)}=(2^{(1/3)})^x$.

FAST METHOD: $b=2^{(1/3)}$.

Q39. Correct Answer: B**STEP-BY-STEP SOLUTION:**

$$A(5)=5,000(1.06)^5=5,000*1.33823\approx\$6,691.$$

$$A(10)=5,000(1.06)^{10}=5,000*1.79085\approx\$8,954.$$

$$\text{Difference}=\$8,954-\$6,691=\$2,263$$

WHY EACH OPTION IS WRONG:

- A) \$1,767: underestimates
- C) \$2,552: arithmetic error
- D) \$3,439: computes $A(10)-A(0)$ instead of $A(10)-A(5)$

TRAP ALERT: Computing $A(10)-A(0)$ (full 10-year growth) instead of $A(10)-A(5)$.

KEY INSIGHT: Calculate each value separately using $(1.06)^5\approx 1.338$ and $(1.06)^{10}\approx 1.791$, then subtract.

FAST METHOD: $8,954-6,691=\$2,263$.

Q40. Correct Answer: 4**STEP-BY-STEP SOLUTION:**

$$b=f(2)/f(1)=36/12=3. \text{ } a=f(0)=f(1)/b=12/3=4$$

TRAP ALERT: Using $f(1)=12$ directly as a .

KEY INSIGHT: Find b from ratio of consecutive values; back-solve for a .

FAST METHOD: $b=3$; $a=12/3=4$.

Q41. Correct Answer: C

STEP-BY-STEP SOLUTION:

I: $n \cdot 3^t$ ✓. II: $n \cdot (1+2)^t = n \cdot 3^t$ ✓ (since $1+2=3$). Both are identical.

WHY EACH OPTION IS WRONG:

- A) Only I: misses that $(1+2)=3$ makes II identical
- B) Only II: I is also correct
- D) Both are correct

TRAP ALERT: Not simplifying $(1+2)=3$ in expression II.

KEY INSIGHT: $1+2=3$, so $(1+2)^t=3^t$. Both expressions are the same.

FAST METHOD: $(1+2)^t=3^t \rightarrow$ I and II are identical.

Q42. Correct Answer: C

STEP-BY-STEP SOLUTION:

As $x \rightarrow -\infty$: $2(3)^x \rightarrow 0 \rightarrow f(x) \rightarrow 0+5=5$. Horizontal asymptote: $y=5$

WHY EACH OPTION IS WRONG:

- A) $y=0$: asymptote of $2(3)^x$ alone, without the $+5$
- B) $y=2$: the coefficient, not asymptote
- D) $y=7=f(0)$: the y -intercept

TRAP ALERT: Giving $y=0$ (bare exponential asymptote) instead of $y=d$ (the vertical shift).

KEY INSIGHT: For $f(x)=ab^x+d$: horizontal asymptote is $y=d$.

FAST METHOD: As $x \rightarrow -\infty$: exponential $\rightarrow 0 \rightarrow y \rightarrow 5$.

Q43. Correct Answer: C

STEP-BY-STEP SOLUTION:

$W(10)=4,000(0.92)^{10}=4,000 \cdot 0.4344 \approx 1,738$ gallons.

Reduction $= 4,000 - 1,738 = 2,262$. Percent $= (2,262/4,000) \cdot 100 \approx 56.6\% \approx 57\%$

WHY EACH OPTION IS WRONG:

- A) 8%: per-minute rate, not total 10-minute reduction
- B) 43%: fraction REMAINING (43.4%), not fraction reduced
- D) 80%: would need $(0.92)^{10} \approx 0.20$, but actually ≈ 0.434

TRAP ALERT: Confusing fraction remaining (43%) with fraction reduced (57%).

KEY INSIGHT: Percent reduced $= 1 - (\text{fraction remaining}) = 1 - (0.92)^{10} \approx 1 - 0.434 = 56.6\%$.

FAST METHOD: $(0.92)^{10} \approx 0.434 \rightarrow$ reduced by $1 - 0.434 = 57\%$.

Q44. Correct Answer: B

STEP-BY-STEP SOLUTION:

$3^{(-x)}$ replaces x with $-x$ in $3^x \rightarrow$ reflection over the y-axis.

WHY EACH OPTION IS WRONG:

- A) Reflection over x-axis: $y=-3^x$
- C) Horizontal shift: $y=3^{(x-h)}$
- D) Different functions: one increases, one decreases

TRAP ALERT: Confusing 'negate input' (y-axis reflection) with 'negate output' (x-axis reflection).

KEY INSIGHT: $f(-x)$: replace x with $-x \rightarrow$ y-axis reflection. $-f(x)$: negate output $\rightarrow$ x-axis reflection.

FAST METHOD: $3^{(-x)}$: $x \rightarrow -x \rightarrow$ y-axis reflection.

Q45. Correct Answer: B

STEP-BY-STEP SOLUTION:

$500(0.5)^{(t/6)}=62.5 \rightarrow (0.5)^{(t/6)}=1/8=(0.5)^3 \rightarrow t/6=3 \rightarrow t=18$

WHY EACH OPTION IS WRONG:

- A) $t=12$: $(0.5)^2=0.25$; $500*0.25=125 \neq 62.5$
- C) $t=24$: $(0.5)^4=0.0625$; $500*0.0625=31.25 \neq 62.5$
- D) $t=30$: $(0.5)^5 \approx 0.031$; result $\approx 15.6 \neq 62.5$

TRAP ALERT: Not recognizing $1/8=(1/2)^3$ to match exponents.

KEY INSIGHT: Set $(0.5)^{(t/6)}=(0.5)^3 \rightarrow$ exponents match $\rightarrow t/6=3$.

FAST METHOD: $62.5/500=1/8=(1/2)^3 \rightarrow t/6=3 \rightarrow t=18$.

Q46. Correct Answer: B

STEP-BY-STEP SOLUTION:

$f(x)=20(1.3)^{(x+k)}=20(1.3)^x*(1.3)^k=(1.3)^k*g(x)$

WHY EACH OPTION IS WRONG:

- A) $g(x)+k$: adds k to output
- C) $k*g(x)$: uses k not $(1.3)^k$
- D) $g(x+k)$: evaluates g at different input

TRAP ALERT: Thinking 'add k to exponent' = 'add k to function value.'

KEY INSIGHT: $b^{(x+k)}=b^k*b^x$. Exponent addition becomes output multiplication.

FAST METHOD: $(1.3)^{(x+k)}=(1.3)^k*(1.3)^x \rightarrow f(x)=(1.3)^k*g(x)$.

Q47. Correct Answer: B**STEP-BY-STEP SOLUTION:**

$ab^2=36$, $ab^5=972$. Divide: $b^3=27 \rightarrow b=3$. Then $a=36/9=4$

WHY EACH OPTION IS WRONG:

- A) $a=3$: $3(9)=27 \neq 36$
- C) $a=9$: $9(9)=81 \neq 36$
- D) $a=12$: $12(9)=108 \neq 36$

TRAP ALERT: Solving for a without eliminating it via the ratio.

KEY INSIGHT: Divide equations to cancel a : $b^3=27$. Then back-solve.

FAST METHOD: $b^3=27 \rightarrow b=3 \rightarrow a=4$.

Q48. Correct Answer: D**STEP-BY-STEP SOLUTION:**

$N(5)=800/(1+3e^{(-2)})$. $e^{(-2)} \approx 0.1353$. Denom $=1+3(0.1353)=1+0.406=1.406$. $N \approx 800/1.406 \approx 569$

WHY EACH OPTION IS WRONG:

- A) $267 \approx 800/3$: denominator at $t=0$
- B) 400: half-maximum, not reached at $t=5$
- C) 491: uses $e^{(-2)} \approx 0.20$ instead of 0.135

TRAP ALERT: Using $e^{(-2)} \approx 0.5$ (confusing with $1/2$) instead of the correct $e^{(-2)} \approx 0.135$.

KEY INSIGHT: $e^{(-2)} \approx 0.135$, not 0.5. Evaluate carefully before computing denominator.

FAST METHOD: $e^{(-2)} \approx 0.135 \rightarrow \text{denom} \approx 1.406 \rightarrow N \approx 569$.

Q49. Correct Answer: B**STEP-BY-STEP SOLUTION:**

$f(x)=32$ when: $16(2)^{(x/4)}=32 \rightarrow 2^{(x/4)}=2 \rightarrow x/4=1 \rightarrow x=4$.

Option B: $f(x)=32 \cdot 2^{((x-4)/4)}$. At $x=4$: $32 \cdot 2^0=32$ ✓. This form shows $x=4$ as the key value.

WHY EACH OPTION IS WRONG:

- A) Same as original; 32 not displayed
- C) Different function
- D) $16^{(x/4)} \neq 16 \cdot 2^{(x/4)}$

TRAP ALERT: Not recognizing the question asks for a form that **DISPLAYS** $x=4$ as a constant.

KEY INSIGHT: When $f(x)=32$: rewrite so 32 appears as the coefficient explicitly.

FAST METHOD: $f(4)=32 \rightarrow$ rewrite as $32 \cdot 2^{((x-4)/4)}$.

Q50. Correct Answer: A**STEP-BY-STEP SOLUTION:**

Annual rate=12%=0.12. Monthly rate $r=0.12/12=0.01$

WHY EACH OPTION IS WRONG:

- B) 0.12: annual rate, not monthly
- C) 1.00=100% monthly: absurd
- D) 1.12: annual growth FACTOR, not monthly rate

TRAP ALERT: Using the annual rate directly as the monthly rate.

KEY INSIGHT: Monthly rate = annual rate / 12.

FAST METHOD: 12%/12 months=1%=0.01.

Q51. Correct Answer: 2.25**STEP-BY-STEP SOLUTION:**

$f(x)=10(1.5)^{(x+2)}=10(1.5)^x(1.5)^2=10(1.5)^x \cdot 2.25=2.25 \cdot g(x)$. So $c=2.25$.

TRAP ALERT: Thinking $c=2$ (the exponent addend) rather than $(1.5)^2$.

KEY INSIGHT: $(1.5)^{(x+2)}=(1.5)^2 \cdot (1.5)^x$. Constant factor= $(1.5)^2=2.25$.

FAST METHOD: $c=(1.5)^2=2.25$.

Q52. Correct Answer: B**STEP-BY-STEP SOLUTION:**

$5 \cdot 4^x - 60 = 0 \rightarrow 4^x = 12$. Since 4^x is strictly increasing and one-to-one, it equals 12 at exactly ONE value of x .

WHY EACH OPTION IS WRONG:

- A) 0: 4^x is always positive and does equal 12
- C) 2: exponential functions are one-to-one
- D) 3: impossible for a strictly monotonic function

TRAP ALERT: Thinking exponential equations can have multiple solutions like quadratics.

KEY INSIGHT: A strictly monotone function crosses any horizontal line at most once.

FAST METHOD: $4^x=12$ has exactly one solution.

Q53. Correct Answer: A

STEP-BY-STEP SOLUTION:

$$f(x) = 7(1.06)^{(x/12)} = 7 * ((1.06)^{(1/12)})^x. \quad b = (1.06)^{(1/12)} \approx 1.00487 \approx 1.005$$

WHY EACH OPTION IS WRONG:

- B) 1.06: annual factor, not per-unit
- C) 1.072: incorrect
- D) 1.12: annual 12% factor

TRAP ALERT: Using annual factor (1.06) as b instead of its 12th root.

KEY INSIGHT: $2^{(x/n)} = (2^{(1/n)})^x$. Take the n th root to convert.

FAST METHOD: $b = (1.06)^{(1/12)} \approx 1.005$.

Q54. Correct Answer: A

STEP-BY-STEP SOLUTION:

$$\text{Set equal: } 400(1.2)^t = 1,600(1.05)^t \rightarrow (1.2/1.05)^t = 4 \rightarrow (1.1429)^t = 4.$$

$$t = \ln(4)/\ln(1.1429) \approx 1.386/0.1335 \approx 10.4 \text{ years} \rightarrow \text{between 10 and 11.}$$

WHY EACH OPTION IS WRONG:

- B) 15-16: too long
- C) 20-21: too long
- D) 25-26: far too long

TRAP ALERT: Not setting up the ratio correctly.

KEY INSIGHT: Divide both sides by $1,600(1.05)^t$ to isolate the ratio.

FAST METHOD: $\ln(4)/\ln(1.143) \approx 10.4 \rightarrow \text{between 10 and 11.}$

Q55. Correct Answer: A

STEP-BY-STEP SOLUTION:

$$f(0) = a = 8. \quad f(3) = 8 * b^3 = 1 \rightarrow b^3 = 1/8 \rightarrow b = 1/2$$

WHY EACH OPTION IS WRONG:

- B) $b = 1/3$: $8 * (1/27) = 8/27 \neq 1$
- C) $b = 1/4$: $8 * (1/64) \neq 1$
- D) $b = 1/8$: $8 * (1/512) \neq 1$

TRAP ALERT: Confusing b with b^3 ; $b^3 = 1/8$ so $b = (1/8)^{(1/3)} = 1/2$.

KEY INSIGHT: $b^3 = 1/8 \rightarrow \text{cube root} \rightarrow b = 1/2$.

FAST METHOD: $b^3 = 1/8 \rightarrow b = 1/2$.

Q56. Correct Answer: 12**STEP-BY-STEP SOLUTION:**

Set equal: $100(2)^{(t/3)} = 400(2)^{(t/6)}$.

Divide: $2^{(t/3-t/6)} = 4 \rightarrow 2^{(t/6)} = 4 = 2^2 \rightarrow t/6 = 2 \rightarrow t = 12$

TRAP ALERT: Arithmetic error: $t/3 - t/6 = t/6$ (not $t/2$ or $t/3$).

KEY INSIGHT: Subtract exponents when dividing same base: $2^{(t/3)}/2^{(t/6)} = 2^{(t/6)}$.

FAST METHOD: $2^{(t/6)} = 4 = 2^2 \rightarrow t = 12$.

Q57. Correct Answer: B**STEP-BY-STEP SOLUTION:**

$a = f(0) = 5$. $f(2) = 5 \cdot 3^{(2b)} = 45 \rightarrow 3^{(2b)} = 9 = 3^2 \rightarrow 2b = 2 \rightarrow b = 1$

WHY EACH OPTION IS WRONG:

A) $b = 1/2$: $5 \cdot 3^1 = 15 \neq 45$

C) $b = 3/2$: $5 \cdot 3^3 = 135 \neq 45$

D) $b = 2$: $5 \cdot 3^4 = 405 \neq 45$

TRAP ALERT: Getting $3^{(2b)} = 9$ and jumping to $b = 3$.

KEY INSIGHT: $3^{(2b)} = 9 = 3^2 \rightarrow$ exponents must match $\rightarrow 2b = 2 \rightarrow b = 1$.

FAST METHOD: $3^{(2b)} = 3^2 \rightarrow b = 1$.

Q58. Correct Answer: B**STEP-BY-STEP SOLUTION:**

$f(0) = c \cdot d^0 = c = 4 \rightarrow c = 4 > 1$. Graph is DECREASING $\rightarrow 0 < d < 1$.

Verify: $4 \cdot d^2 = 1 \rightarrow d = 1/2$ (satisfies $0 < d < 1$). ✓

WHY EACH OPTION IS WRONG:

A) $c < 1$: contradicts y-intercept $= 4 > 1$

C) $d > 1$: would be increasing (contradicts graph)

D) $c < 0$: function would be negative

TRAP ALERT: Confusing the y-intercept c ($=4$) with the base d .

KEY INSIGHT: $c = f(0) =$ y-intercept. Decreasing $\rightarrow 0 < d < 1$. Check both conditions.

FAST METHOD: $c = 4 > 1$; decreasing $\rightarrow 0 < d < 1 \rightarrow$ Answer B.

Q59. Correct Answer: A

STEP-BY-STEP SOLUTION:

Asymptote $y=3 \rightarrow d=3$. $f(x)=a \cdot b^x+3$.

$f(0)=a+3=7 \rightarrow a=4$. $f(1)=4b+3=11 \rightarrow 4b=8 \rightarrow b=2$

WHY EACH OPTION IS WRONG:

B) $b=3$: $4(3)+3=15 \neq 11$

C) $b=4$: $4(4)+3=19 \neq 11$

D) $b=8$: $4(8)+3=35 \neq 11$

TRAP ALERT: Using $f(0)=7$ as a directly (forgetting to subtract $d=3$).

KEY INSIGHT: Subtract asymptote: $a=f(0)-d=7-3=4$.

FAST METHOD: $a=4$; $b=(11-3)/4=2$.

Q60. Correct Answer: 30.375

STEP-BY-STEP SOLUTION:

$f(n+1)/f(n)=1.5 \rightarrow$ geometric sequence, ratio=1.5.

$f(1)=6$, $f(2)=9$, $f(3)=13.5$, $f(4)=20.25$, $f(5)=30.375$

Or: $f(5)=6 \cdot (1.5)^4=6 \cdot 5.0625=30.375$

TRAP ALERT: Adding 1.5 each time (arithmetic) instead of multiplying (geometric).

KEY INSIGHT: Constant ratio $f(n+1)/f(n)=1.5$ defines a geometric (exponential) sequence.

FAST METHOD: $f(5)=6 \cdot (1.5)^4=30.375$.

TOPIC SUMMARY — Exponential Functions - Growth and Decay

Common Mistakes:

1. Using the growth/decay RATE as the base (e.g., 0.06) instead of the growth FACTOR ($1.06=1+0.06$).
2. Exponent trap: adding k to the exponent multiplies the function by b^k (it does NOT add k to the output).
3. Half-life error: the exponent is t/T (number of periods elapsed), not t itself.

Must Remember:

1. Growth: $f(t)=a(1+r)^t$. Decay: $f(t)=a(1-r)^t$. The base is always $(1 \pm \text{rate})$.
2. Half-life T : $Q(t)=Q_0(0.5)^{(t/T)}$. Exponent counts how many half-lives have passed.
3. Horizontal asymptote of $f(x)=a \cdot b^x + d$ is $y=d$ (the additive constant, not zero).

Expected on DSAT:

2-3 exponential questions per Digital SAT (1-2 in Module 1, 1-2 in Module 2). Hard module tests converting forms, logistic growth, or comparing two exponential models.

